Conditional GAN for Sketch-to-Face Translation with Perceptual and L1 Reconstruction Losses

Huzaifa Rehman (i21-2694)
Department of AI and Data Science — 02 Nov 2025

Abstract—We implement a conditional GAN (cGAN) that maps edge/sketch inputs to photorealistic faces. Our generator follows a U-Net with skip connections; the discriminator is a PatchGAN operating on (sketch, face) pairs. The objective combines adversarial loss with L1 reconstruction and a
VGG-based perceptual term. We train on a Google Drive dataset of paired sketches and faces, and report training curves, qualitative pairs, and validation SSIM/PSNR.
Index Terms— Conditional GAN, Pix2Pix, Sketch-to-Face, Perceptual Loss, PatchGAN, U-Net

I. INTRODUCTION

Image-to-image translation learns a mapping between two visual domains given paired supervision. For sketch→face, paired data provide a strong geometric prior (edges) but lack texture; the cGAN prior encourages photo realism. We adopt a Pix2Pix-like framework with additional perceptual guidance to synthesize sharper details while preserving structure.

II. DATASET AND PREPROCESSING

The dataset is provided as a ZIP on Google Drive containing paired sketches and faces. During loading we form tuples (x_sketch, y_face). Preprocessing converts images to RGB float tensors in [−1,1], resizes to 256×256, and applies random horizontal flip and slight affine jitter. Batches are balanced and shuffled each epoch.

III. MODEL

Generator: U-Net encoder–decoder with skip connections (down blocks: Conv–BN–LeakyReLU; up blocks: ConvTranspose–BN–ReLU with dropout in deep layers).
Discriminator: 70×70 PatchGAN on the concatenated [x_sketch, y] input. **Loss:** adversarial (BCE), L1 reconstruction (λ_{L1}), and VGG-perceptual (λ_P). The total generator objective is: $L_{GAN} = L_{L1} + \lambda_P \cdot \|y - G(x)\| + \lambda_P \cdot \|\phi(y) - \phi(G(x))\|$.

IV. TRAINING

Optimization uses Adam (β=0.5, β=0.999) with a cosine learning rate schedule. Mixed precision is enabled on T4. We log losses per N steps and save sample grids. Validation uses SSIM and PSNR. Qualitative inspection is done via side-by-side panels of input sketch and generated face.

Metric	Value
Min loss_G ↓	34.5017
Min loss_D ↓	0.0254
Val SSIM ↑	0.2806
Val PSNR ↑ (dB)	11.55

Table I. Training summary and validation metrics (from logs).

V. RESULTS

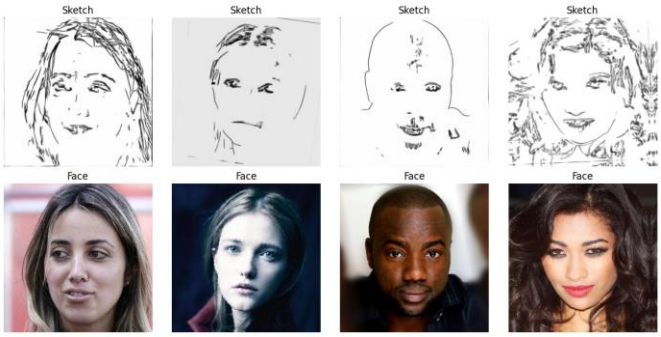

Fig. 1. Paired samples from the dataset (sketch→face).

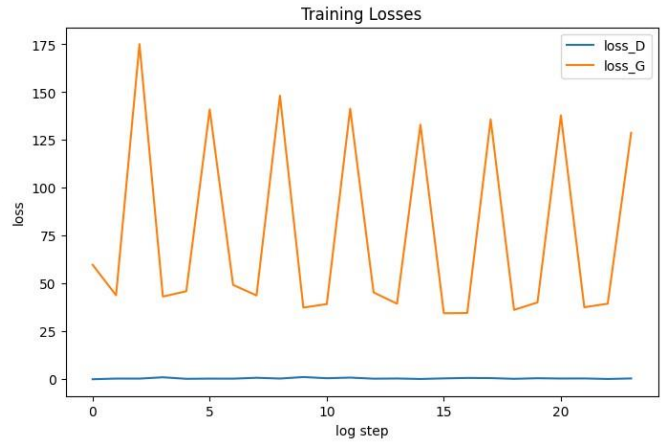

Figure 1 shows examples from the dataset (paired sketch/face).
Fig. 3. Training losses for discriminator and generator (log steps).
Figure 2 illustrates an input sketch and a generated face. Figure 3 plots training losses for the discriminator and generator across

logging steps; the generator's curve reflects the weighted sum of adversarial, L1, and perceptual terms. Table I summarizes the final and best losses together with validation SSIM/PSNR. While the model reconstructs global structure, texture and color bleeding remain, indicating room for architectural tuning and increased data quality.

VI. DISCUSSION

Perceptual loss helps stabilize training and improves identity features (eyes/mouth), but over-weighting it can introduce stylization artifacts. PatchGAN focuses on local realism; adding multi-scale discriminators and a feature matching term typically reduces checkerboard artifacts. Higher resolution crops (e.g., 512^2 with coarse to fine generators) and stronger augmentation can further improve results.

VII. CONCLUSION

A Pix2Pixstyle cGAN with L1+perceptual objectives produces reasonable sketch→face translations on limited paired data. Future work includes multiscale discriminators, SPADE style conditioning, or diffusion based translation for higher fidelity.

VIII. PROMPTS

Prompts/automation used: mount Drive and unzip; build paired dataset loader; U-Net generator + Patch GAN; train with Adam and cosine LR; log losses; export sample panels and curves; compute SSIM/PSNR;

REFERENCES

[1] Isola et al., "Image-to-Image Translation with Conditional Adversarial Networks," CVPR 2017.

[2] Ledig et al., "Photo-Realistic Single Image Super-Resolution Using a GAN," CVPR 2017 (perceptual/VGG).

[3] Johnson et al., "Perceptual Losses for Real-Time Style Transfer and Super-Resolution," ECCV 2016.